

CODECHECK certificate 2025-023

<https://codecheck.org.uk/register/certs/2025-023/>



Table 1: CODECHECK summary

Item	Value
Title	<i>A calibrated optogenetic toolbox of stable zebrafish opsin lines</i>
Author(s)	Paride Antinucci  , Adna Dumitrescu  , Charlotte Deleuze, Holly J Morley  , Kristie Leung, Tom Hagley, Fumi Kubo, Herwig Baier  , Isaac H Bianco  , Claire Wyart 
Reference	https://doi.org/10.7554/eLife.54937
Repository	https://github.com/codecheckers/certificate-2025-023
Codechecker(s)	Linus Dexter Hackel 
Date of check	2026-01-15
Summary	Figures 4, 5, 8 and 9 could be reproduced partially. The provided scripts produce parts of the papers figures but there is always only one produced graph which matches a graph in the related figure in the paper. It is unclear why, or how that graph is selected and why the rest of the corresponding figure is missing. However, some produced figures look completely different to those in the paper, e.g., “figures/figure4_1.pdf”, whose graphs aren’t included in Figure 4 of the paper at all. There was no documentation how Figures 1, 2, 3, 6 and 10 were created.

Table 2: Summary of output files generated

File	Comment	Size (b)
figure4_1.pdf	Manuscript Figure 4 (Trace: 2019_03_16_0000, Opsin: Cheriff)	16548
figure4_2.pdf	Manuscript Figure 4 (Trace: 18n270027_1, Opsin: Chromson)	82263
figure4_3.pdf	Manuscript Figure 4 (Trace: 2019_03_19_0038, Opsin: CoChR)	63693
figure5_1.pdf	Manuscript Figure 5 (Trace: 2019_03_19_0055, Opsin: GtACR1)	78091
figure5_1.pdf	Manuscript Figure 5 (Trace: 18d130007_5, Opsin: Chromson)	78091
figure8_1.pdf	Manuscript Figure 8 (Trace: 183060053_1, Opsin: NpHR)	235116
figure8_2.pdf	Manuscript Figure 8 (Trace: 2019_08_01_0064, Opsin: GtACR1)	213717
figure9_1.pdf	Manuscript Figure 9 (Trace: 186280022_1, Opsin: NpHR)	245474
figure9_2.pdf	Manuscript Figure 9 (Trace: 2019_02_25_0035, Opsin: GtACR1)	248817

File	Comment	Size (b)
figure9_3.pdf	Manuscript Figure 9 (Trace: 2019_08_01_0067, Opsin: GtACR1)	179181
figure9_4.pdf	Manuscript Figure 9 (Trace: 189040010_1, Opsin: NpHR)	94949
figure9_5.pdf	Manuscript Figure 9 (Trace: 2019_01_25_0007, Opsin: GtACR1)	121586

Summary

Figures 4, 5, 8 and 9 could be reproduced partially. The provided scripts produce parts of the papers figures but there is always only one produced graph which matches a graph in the related figure in the paper. It is unclear why, or how that graph is selected and why the rest of the corresponding figure is missing. However, some produced figures look completely different to those in the paper, e.g., “figures/figure4_1.pdf”, whose graphs aren’t included in Figure 4 of the paper at all. There was no documentation how Figures 1, 2, 3, 6 and 10 where created.

CODECHECKER notes

Environment

Setting up the environment took a bit of time, as older versions of Python and older Dependencies needed to be properly installed, but it is all very well documented in the **README** file what dependencies need to be installed and with which version.

Since I don’t have a graphical UI, but only a Linux command line, opening a new window for the plot results of Matplotlib didn’t work. To solve this, I decided to put the following sequence at the end of each script, so the created figure is automatically saved to the **outputs/figures/** directory with the desired name.

```
figure_number = 9

figure_id = int(input('Please enter the Figure ID.\n\n'))

plt.savefig(f"figures/figure{figure_number}_{figure_id}.pdf")
```

Script Errors

In the script [Excitatory_Opsin_Current_Clamp.py](#) the filepath for the trace contained a typo in line 452. It was: '[Analysis_output/Single_trace_data/CC_excitatory/2019_03_19_0055.csv](#)' but it should have been '[Analysis_output/Single_Trace_data/CC_excitatory/2019_03_19_0055.csv](#)'. So instead of **Trace** the url included **trace**. This is just a small error, but it still took me some minutes in figuring out, where the **FileNotFoundError** could be coming from, as the file appeared to be there.

Similar to the first error, in the script [Inhibitory_Opsin_Current_Clamp.py](#) the filepath for the CC inhibitory opsin master sheet wasn’t correct and needed to be changed in the lines 400, 417, 423. Here the problem was, that it was the full file path to the authors private directory, so it needed to be changed to my directory. My recommendation is, to only use relative file paths, such that these errors can be avoided. Different to the first error though, the file didn’t exist entirely, so it had to be newly created with the same **csv-Header** as the file '[Analysis_output/VC_inhibitory_opsin_master.csv](#)'. After these two fixes, the script worked perfectly.

In the script [Inhibitory_Opsin_CC_Long_AP_Inhibit.py](#) the line 424 had to be changed from an array of the length 7 to an array of the length 2, as this was what was given in the documentation and also what the code afterwards expected. The code was therefore changed from:

```
LED_max_V_user = [input('pulse1: \n'), input('pulse2: \n'), input('pulse3: \n'), input('pulse4: \n'),
input('pulse5: \n'), input('pulse6: \n'), input('pulse7: \n')]
```

to

```
LED_max_V_user = [input('pulse_1: \n'), input('pulse2: \n')]
```

Recommendations to the authors

Firstly, please add documentation on how to reproduce the Figures 1, 2, 3, 6 and 10 and on how to reproduce the missing pieces of the Figures 4, 5, 8 and 9. Furthermore, make sure to use relative file paths everywhere, so the scripts can be executed from different machines without modifications (see [Script Errors](#)). Finally, I have to say that apart from the small errors and the missing documentation, the remaining documentation was really well done and the reproduction of the figures which worked was really easy and straight forward.

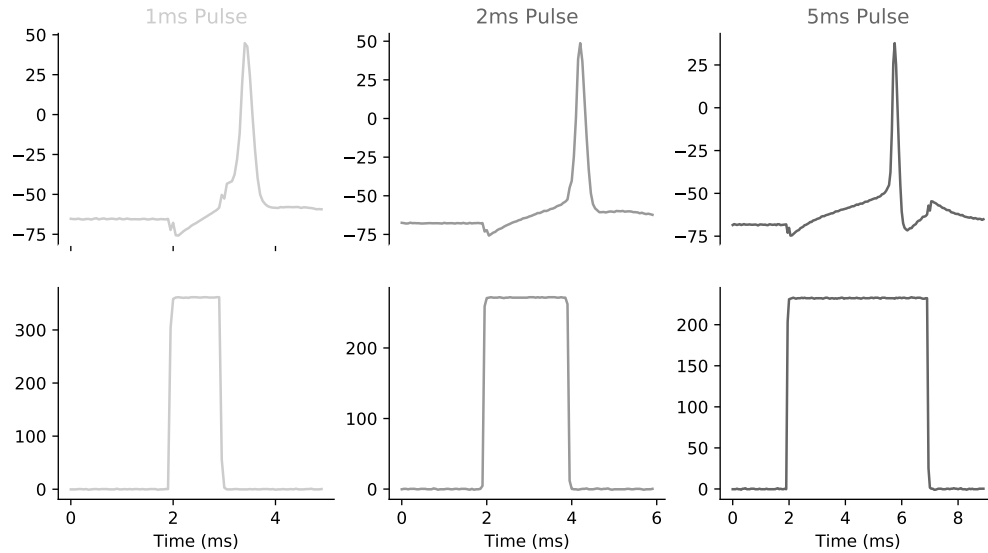
Manifest files

CSV files

Figures

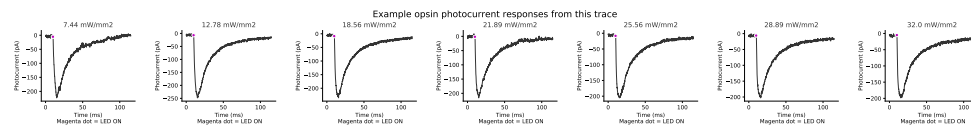
figures/figure4_1.pdf

Author comment: *Manuscript Figure 4 (Trace: 2019_03_16_0000, Opsin: Cheriff)*



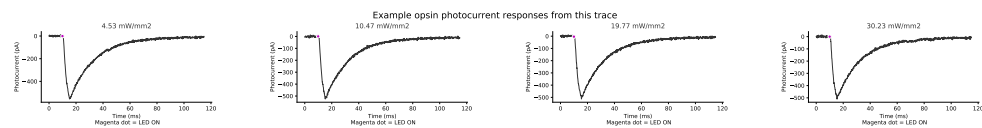
figures/figure4_2.pdf

Author comment: *Manuscript Figure 4 (Trace: 18n270027_1, Opsin: Chrimson)*



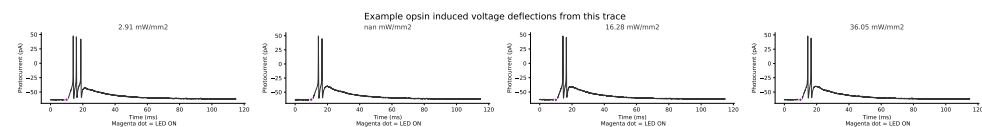
figures/figure4_3.pdf

Author comment: *Manuscript Figure 4 (Trace: 2019_03_19_0038, Opsin: CoChR)*



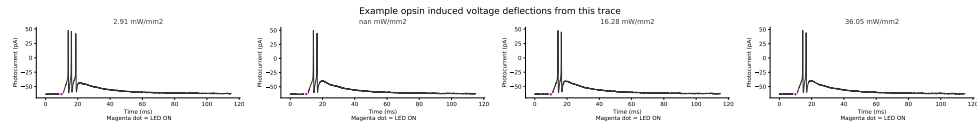
figures/figure5_1.pdf

Author comment: *Manuscript Figure 5 (Trace: 2019_03_19_0055, Opsin: GtACR1)*



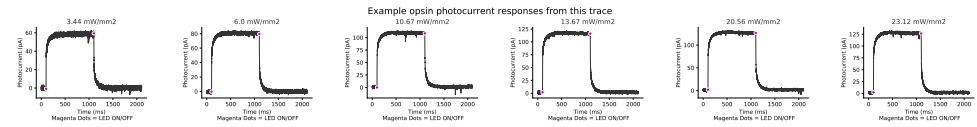
figures/figure5_1.pdf

Author comment: *Manuscript Figure 5 (Trace: 18d130007_5, Opsin: Chrimson)*



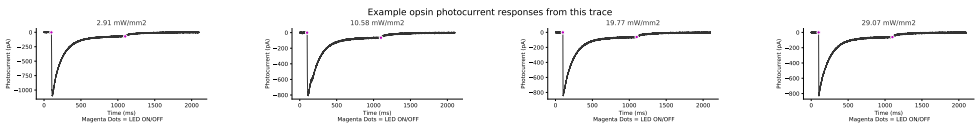
figures/figure8_1.pdf

Author comment: *Manuscript Figure 8 (Trace: 183060053_1, Opsin: NpHR)*



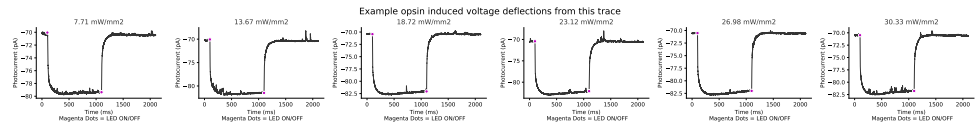
figures/figure8_2.pdf

Author comment: *Manuscript Figure 8 (Trace: 2019_08_01_0064, Opsin: GtACR1)*



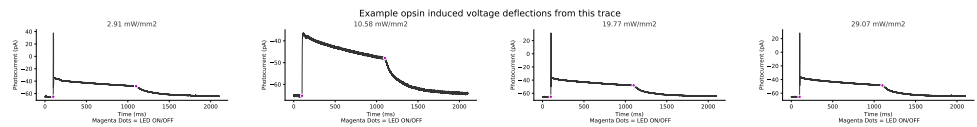
figures/figure9_1.pdf

Author comment: *Manuscript Figure 9 (Trace: 186280022_1, Opsin: NpHR)*



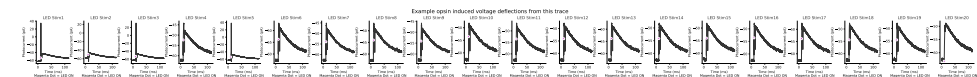
figures/figure9_2.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_02_25_0035, Opsin: GtACR1)*



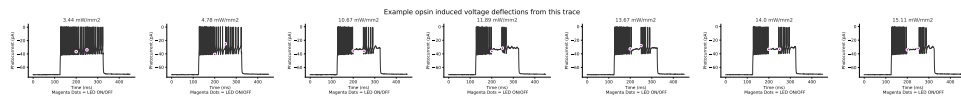
figures/figure9_3.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_08_01_0067, Opsin: GtACR1)*



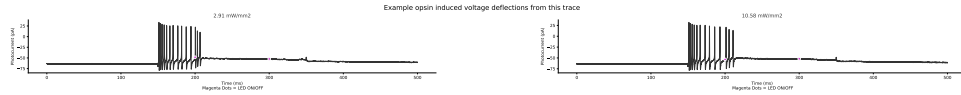
figures/figure9_4.pdf

Author comment: *Manuscript Figure 9 (Trace: 189040010_1, Opsin: NpHR)*



figures/figure9_5.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_01_25_0007, Opsin: GtACR1)*



Acknowledgements

Dr. Claire Wyart tried to help in contacting the author of the code to plot the data on electrophysiology ([Adna Dumitrescu](#)) and the author responsible for the figures on behavioral responses to optogenetic stimulations ([Paride Antinucci](#)). However neither of them responded when I contacted them.

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Citing this document

Linus Dexter Hackel (2026). CODECHECK Certificate 2025-023. Zenodo. <https://codecheck.org.uk/register/certs/2025-023/>

About CODECHECK

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About this document

This document was created using [codecheck-py](#) (a Python-base template for creating [CODECHECK](#) certificates). The CODECHECK details are filled into a [jupyter notebook](#) which is then converted into Markdown via [nbconvert](#). Afterwards it gets converted into [Typst](#) using [cmarker](#) and then into PDF using Typst. `sh notebook_to_pdf.sh` will regenerate the report file.

```
import session_info2 as si
si.session_info(os=True, cpu=True, gpu=True, dependencies=True)
```

```
wcwidth 0.2.14
tornado 6.5.3
Pygments 2.19.2
pyzmq 27.1.0
packaging 25.0
psutil 7.1.3
ipython 9.8.0
urllib3 2.6.1
PyYAML 6.0.3
Brotli 1.2.0
prompt_toolkit 3.0.52
jupyter_client 8.7.0
debugpy 1.8.17
jupyter_core 5.9.1
decorator 5.2.1
traitlets 5.14.3
asttokens 3.0.1
stack_data 0.6.3
platformdirs 4.5.1
comm 0.2.3
idna 3.11
setuptools 80.9.0
jedi 0.19.2
colorama 0.4.6
pure_eval 0.2.3
PySocks 1.7.1
pandas 3.0.0
python-dateutil 2.9.0.post0
ipykernel 7.1.0
charset-normalizer 3.4.4
```

```
requests 2.32.5
parso 0.8.5
executing 2.2.1
session-info 0.3
pytz 2025.2
six 1.17.0
certifi 2026.1.4 (2026.01.04)
numpy 2.3.5
----
Python 3.14.2 | packaged by conda-forge | (main, Dec 6 2025, 11:21:58) [GCC 14.3.0]
OS Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.35
CPU 16 logical CPU cores, x86_64
GPU No GPU found
Updated 2026-03-23 14:00
```

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